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Research Letter

Digital phenotyping and ASCVD risk: An exploratory cross-sectional analysis using online behavioral data

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ABSTRACT

A retrospective, exploratory cross-sectional analysis exploring whether social media data is associated with cardiovascular disease (CVD) risk beyond traditional clinical models. While social media data may capture behavioral and social markers relevant to CVD, their associations with CVD risk remains uncertain.

1. Introduction

Cardiovascular disease (CVD) remains the leading cause of death in the United States [1]. To guide prevention and treatment efforts, clinicians frequently rely on risk prediction tools such as Framingham Risk Score or atherosclerotic cardiovascular disease (ASCVD) to estimate a patient's likelihood of experiencing a cardiovascular event. Although clinicians routinely consider social and behavioral factors like history of exercise, access to healthy foods, etc. when assessing CVD risk, prediction tools and models used datasets that did not include these variables [1,2]. Previous research found that data from social media platforms and wearable devices may sharpen these risk predictions [3]. As social media and digital platforms are increasingly used to document lifestyle and health behaviors. Data from digital sources may provide a window into manifestations of novel risk factors and potentially a better characterization of existing CVD risk factors [4]. For example, users can explicitly share health habits such as "I just ran 10 miles at the gym" or "I just ate three brownies" or subtle markers of health like changes in language or sentence structure reflecting mental status change, cognitive decline, social isolation [4]. Language expressed on social media platforms can reflect upstream psychosocial and behavioral factors that contribute to CVD risk, even when disease is not mentioned directly [3–6]. Incorporating clinical, demographic, social and behavioral markers from social media platforms and wearable devices may augment conventional risk models by offering a more comprehensive understanding of individuals' health, lifestyle, and well-being patterns [4,7].

Prior research found that predictive models that include social media data such as language extracted from Facebook wall posts can provide important contextual information [8,9] while other work found that language extracted from Facebook wall posts sometimes outperform predictions compared with traditional clinical variables such as age, sex, and race [3]. However, it remains unclear whether social media data, like Facebook wall post language can meaningfully enhance CVD risk assessment.

2. Methods

This is a retrospective cross-sectional study that follows the STROBE

(Strengthening the Reporting of Observational Studies in Epidemiology) guideline for reporting observational studies. It was approved by the Institutional Review Board at the University of Pennsylvania (IRB# 833699) and registered on clinicaltrials.gov (NCT04574882).

2.1. Participants

We identified a retrospective cohort of individuals between the ages of 30–74 with and without coronary heart disease (CHD) (ICD 9:414.0, ICD 10: 125) who received care in the University of Pennsylvania Health System, were able to provide informed consent, and willing to share Facebook data from September 2020 to February 2025. Patients were contacted digitally (e.g., text message or email outreach) or in-person during an outpatient visit. Patients provided informed consent, shared historical Facebook wall posts [4,8,9] (all posts before and after CHD clinical diagnosis), and completed a survey that includes demographics, diet, physical activity, smoking and tobacco use, and digital data use and behaviors. Data were stored in a Health Insurance Portability and Accountability Act (HIPAA)-compliant secure server and participants could elect to discontinue sharing data at any time point (Online Supplement: eTable 1 and eFigure 2). As study compensation, enrolled patients were entered into a raffle (1/500 chance to win an iPad) and received a \$50 gift card. Additional information about the enrollment and data collection processes are published elsewhere [10].

2.2. Statistical analyses

We used descriptive statistics to summarize patient demographics and Facebook wall posts. To analyze language data from Facebook wall posts, we employed the Differential Language Analysis Toolkit (DLATK), a Python-based machine learning framework designed for text analysis [10]. We extracted unigram features from participants' Facebook wall posts and supplemented this with psycholinguistic features from the 2022 Linguistic Inquiry and Word Count (LIWC) dictionary [11].

For each participant, we included all available Facebook wall posts shared from the beginning of their account history through the date of data collection, regardless of whether they occurred before or after a CHD clinical diagnosis. We first assessed the association between

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linguistic features (unigrams, LIWC categories, LDA topics) and cardiovascular case status (CHD presence vs absence) using Pearson correlation and logistic regression. Unigrams refer to single-word tokens that capture the frequency of individual terms, including common words, colloquialisms, and domain-specific jargon, within users' Facebook posts. Additionally, we applied latent dirichlet allocation (LDA), a systematic approach to identifying text-based themes, to identify 200 thematic clusters of co-occurring words, referred to as "topics." [12] Separate, simple logistic regression models for each individual feature (unigram, LIWC category, or LDA topic), with Pearson correlation coefficients calculated to evaluate whether language-derived features could significantly identify case status. Each language-derived feature (unigram, LIWC category, or LDA topic) was encoded as a normalized frequency count per user.

To evaluate the association between language data from Facebook wall posts and ASCVD risk scores, we extracted risk scores automatically generated in the electronic health record (EHR). We conducted both regression and classification analyses. Pearson correlation coefficients were used to measure how well language features correlated with continuous ASCVD risk scores. For classification tasks, we assessed the area under the receiver operating characteristic curve (AUC) for models associating both binary risk status (high versus low ASCVD risk, defined as $\geq 10\%$ versus $<10\%$) and established ASCVD risk categories ($<5\%$, $5\% - 7.4\%$, $7.5\% - 9.9\%$, and $\geq 10\%$) [1,2].

Three logistic regression models were compared: one using language features alone, one using demographic variables alone (age, gender, and race), and one combining both language and demographic features. To evaluate associations, we employed stratified 10-fold cross-validation: in each iteration, the model was trained on 90% of the data and tested on the held-out 10%, with this process repeated across all folds. The reported performance metrics reflect the average cross-validated results across these folds. All analyses were conducted using Python 3.10.

3. Results

3.1. Study sample

Over the course of 4.5 years, this study contacted over 60,000 individuals via email outreach, 7636 responded to outreach, 2554 were eligible, 781 completed enrollment, and 527 had sufficient Facebook data and were included in the analysis (Online Supplement: eFigure 1). Participants had a mean age 57.5 years, 287 (54.46%) were female, 292 (55.41%) self-identified as White, 151 (28.65%) as Black, and 194 (36.81%) had a history of CHD while 333 (63.19%) served as controls (1:2 ratio). In total, 1,588,681 Facebook wall posts were analyzed, with a mean of 3014 posts per participant (Table 1).

Table 1
Participant demographics and digital data volume.

	Count	Percentage
Total Participants	527	
Mean Age (SD)	57.5 (12.6)	
Sex		
Female	287	54.46
Male	168	31.88
Unknown	72	13.66
Race		
Asian	12	2.28
American Indian or Alaska Native	2	3.8
Black or African American	151	28.65
White	292	55.41
Unknown	70	13.28
Case/Control		
Case (CHD history)	194	36.81
Control (no CHD history)	333	63.19
Total # of Facebook Posts	1,588,681 (mean: 3014)	

3.2. Digital language analysis: case vs. control

Language features derived from Facebook posts were examined to assess their association with case status (CHD presence or absence). None of the LIWC categories were significantly associated with case status after Bonferroni correction. Similarly, no individual unigrams or LDA-derived topics (0 out of 200) demonstrated statistically significant correlations with case status (Online Supplement).

3.3. Modeling of case status

We evaluated the associations between language-based models for distinguishing cases from controls. Logistic regression models were trained using features derived from LIWC categories, unigrams, and LDA topics. The AUC was 0.54 for LIWC, 0.52 for unigrams, and 0.52 for LDA topics, indicating limited discriminatory power of language features alone in correlating with case status.

3.4. ASCVD risk score analysis

A total of 338 participants had available ASCVD risk score data in their EHR and were included in this analysis (Online Supplement: eTable 1). Within this sample, 119 participants (35.21%) were classified as cases and 219 (64.79%) as controls. Based on ASCVD risk scores, 61.2% of participants were low risk ($<10\%$) and 38.8% high risk ($\geq 10\%$): 39.9% had scores $<5\%$, 10.4% between 5 and 7.4%, 11.0% between 7.5 and 9.9%, and 38.8% $\geq 10\%$.

3.5. Associations and classification of ASCVD risk score

Pearson correlation coefficients between these features and the continuous ASCVD risk score were low: -0.078 for LIWC, 0.117 for unigrams, and 0.205 for LDA topics. For binary classification of high ($\geq 10\%$) versus low ($<10\%$) risk, models using only language features yielded AUCs of 0.55 (LIWC), 0.61 (unigrams), and 0.64 (LDA). When demographic controls (age, sex, race) were added to the models, AUCs improved to 0.79 (LIWC), 0.61 (unigrams), and 0.81 (LDA). A model using only demographic controls achieved an AUC of 0.85. For classification across ASCVD risk categories ($<5\%$, $5\% - 7.4\%$, $7.5\% - 9.9\%$, and $\geq 10\%$), language-only models achieved AUCs of 0.67 (LIWC), 0.71 (unigrams), and 0.67 (LDA). When adding demographic factors, AUCs increased to 0.80 (LIWC), 0.74 (unigrams), and 0.82 (LDA). The demographic-only model reached an AUC of 0.87 (Online Supplement: eTable 2).

4. Discussion

Despite negative findings, this study found that for binary classification of ASCVD risk (high $\geq 10\%$ vs. low $<10\%$), models based solely on language features demonstrated modest performance but when we incorporated demographic variables, such as age, gender, and race, model performance significantly improved (AUCs increased to 0.79 for LIWC, remained at 0.61 for unigrams, and 0.81 for LDA). These findings suggest that while language-based features show some discriminatory signal, particularly when combined with demographic data, traditional clinical risk factors remain more strongly associated with ASCVD risk score.

There are several reasons why our analysis was not statistically significant. We did not account for variability in Facebook posting behavior, frequency, length, or types of content. We included all available Facebook wall posts from account inception through data collection to maximize data volume and language signal, recognizing that our primary aim was to explore associations between digital language patterns and cardiovascular risk, not to establish causal or temporally prior predictors. There may be a temporal relationship between diagnosis and Facebook wall post which could introduce bias or inverse causality.

Furthermore, eFigure 1 demonstrates that most individuals did not respond to the initial outreach and less successfully shared digital data, limiting our power to conduct meaningful analyses.^{6,13–165,17,18} Our study was relatively small ($n = 527$) and the demographic composition was primarily female and White. Further, there were more controls than cases (333 vs. 194) and we only included study participants with an ASCVD risk score documented within a single health system's EHR, which may have excluded participants whose scores were recorded in other health systems, who did not have a score calculated within our system, or who were missing risk score parameters. Furthermore, the cross-sectional design limits causal inference; notably, digital postings may have occurred after the CHD onset and diagnosis. Future work may benefit from modeling language trajectories around diagnosis more explicitly.

Despite these limitations, this study demonstrates that 57% of consented participants shared Facebook wall posts, provided sufficient data, and granted EHR access to support CVD research. Although we did not observe such associations, digital language can reflect psychosocial and behavioral factors linked to cardiovascular risk, which provided the conceptual rationale for testing these signals in this study.^[6,13–16] Previous research has demonstrated more promise in mental health research, where posts are inherently more emotionally expressive.^[5,17,18] Prior studies have shown that patients who choose to share social media or other digital data for research typically do so out of altruistic motivation and trust in academic institutions, while also expressing understandable caution about privacy and data interpretation^[7,19]. These patterns likely shaped participation in the present study and remain important considerations for future digital health research. While collecting and summarizing digital data in CVD correlation was feasible, it was logistically challenging and did not yield statistically significant associations. However, digital data at a population-level holds promise due to its broad reach and overall accessibility. Continued exploration of social media data may help refine predictive models and create new opportunities to better understand CVD at both the individual- and population-levels.

Ethical

It was approved by the Institutional Review Board at the University of Pennsylvania (IRB# 833699).

CRediT authorship contribution statement

Lauren Southwick: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology. **Neil K.R. Sehgal:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Dena Torrente:** Project administration. **Veronica Murgulescu:** Project administration. **Devon Schroeder:** Project administration. **David A. Asch:** Writing – review & editing, Supervision, Funding acquisition. **Lyle Ungar:** Writing – review & editing, Supervision, Methodology. **Nandita Mitra:** Writing – review & editing, Supervision, Methodology. **Peter Groenveld:** Writing – review & editing, Supervision. **Stephen E. Kimmel:** Writing – review & editing, Supervision. **Gary E. Weissman:** Writing – review & editing, Supervision. **Sharath Chandra Guntuku:** Writing – review & editing, Supervision, Software, Methodology, Formal analysis, Data curation. **Raina M. Merchant:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

Artificial Intelligence (AI) application was used to assist with the data collection, writing or review of this manuscript.

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Declaration of competing interest

Lauren Southwick, MPH has nothing to disclose
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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ahjo.2026.100717>.

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